



# YOLO Ensemble for UAV-based Multispectral Defect Detection in Wind Turbine Components

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## Deep Learning

YOLOv8 + specialized thermal model

## UAV Inspection

Multispectral data collection

## Wind Turbines

Defect detection in **blades, towers, motors**

## Performance

mAP@.5: 0.93 | F1-score: 0.90

# Abstract

## Research Overview

**0.93**  
mAP@.5

**0.90**  
F1-Score

Enhanced defect detection in wind turbine components through **YOLO ensemble** integrating visible and thermal channels from UAV inspections. Combines general-purpose YOLOv8 with specialized thermal model using **bounding box fusion algorithm**, outperforming standalone models.



### UAV Inspection

Advanced sensors enable monitoring of wind turbine blades, towers, and critical components



### Multispectral Fusion

Integration of RGB and thermal imagery for enhanced defect visibility



### YOLO Ensemble

Sophisticated fusion algorithm improves detection of both visual and thermal defects

# Introduction

## ⚠ Inspection Challenges

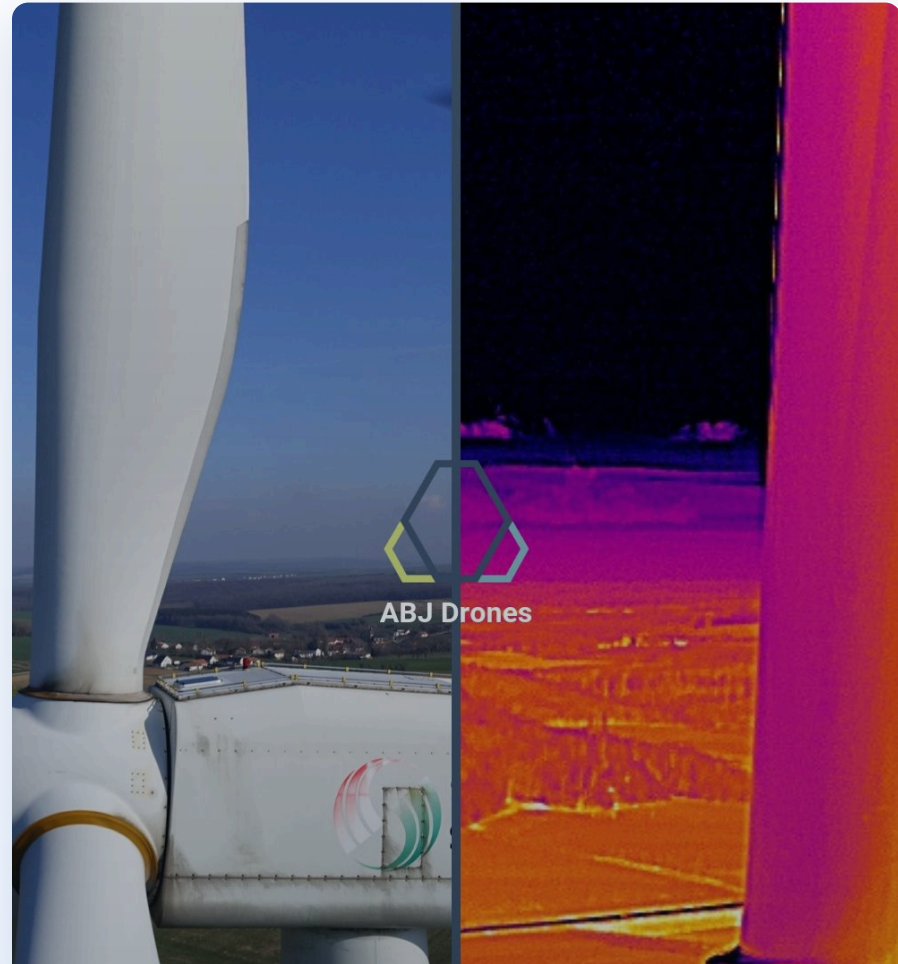
- Minor defects → **economic & safety risks**
- Timely maintenance → **efficiency & longevity**
- Complex geometries & remote locations

## 🚁 UAV-Based Inspection

- **High-resolution optical sensors** & thermal cameras
- Efficient data collection from multiple angles
- Captures defects invisible in RGB images

## 🔧 Data Processing

- Large data volumes → **real-time processing**
- Environmental conditions affect quality



# Related Work



## Sensing Evolution

- > **RGB sensors** with basic ML
- > **Thermal cameras** for temperature anomalies
- > **Multispectral sensors** for early corrosion
- > **5G networks** & edge computing



## Algorithm Evolution

- > **Hand-crafted features** classifiers
- > **Deep learning** for defect patterns
- > **YOLO** for real-time detection
- > **Faster/Cascade R-CNN** for precision



## Fusion Evolution

- > **Simple averaging** of RGB & IR data
- > **CNN-based fusion** learning strategies
- > **Ensemble methods** combining detectors
- > **Research gap** in multispectral ensembles



## State-of-the-Art



**Multispectral data** with advanced CNNs for reliable detection



**Ensemble techniques** balance speed & accuracy for real-time inspection



**RGB-IR fusion** effectively reduces missed defects

# Methodology Overview

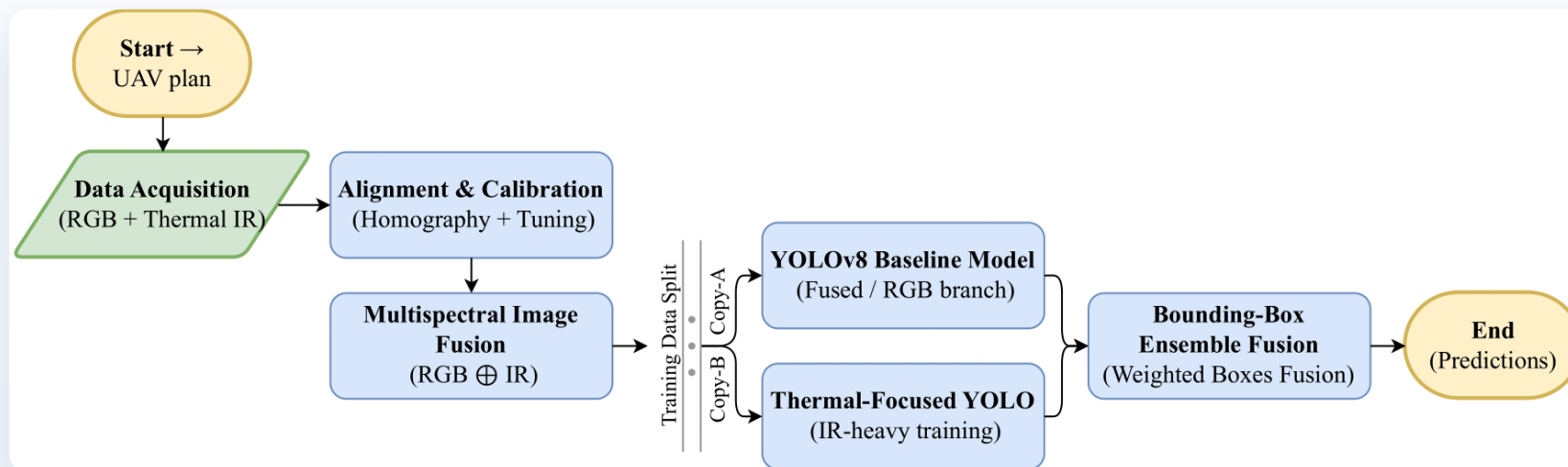
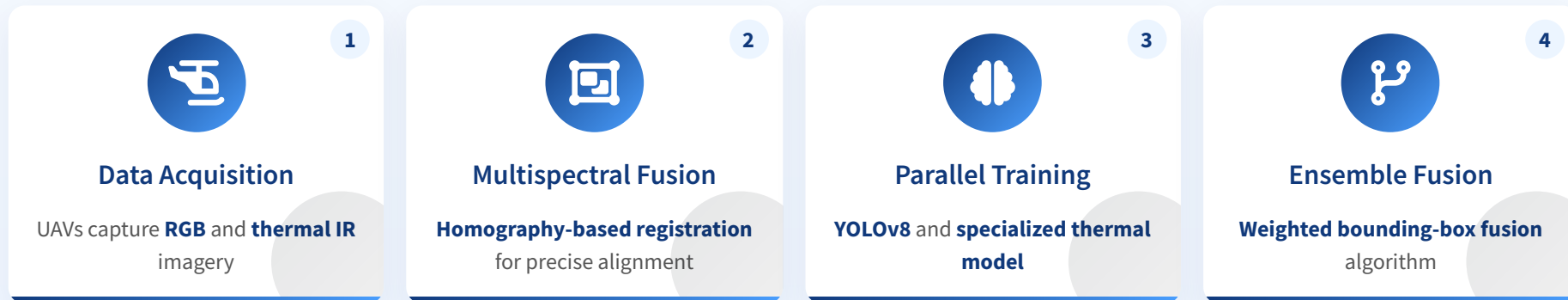


Figure 1: General workflow of our proposed ensemble approach



# Data Collection and Fusion



## RGB Channel

- ✓ **High-resolution** optical images (4K/6K)
- ✓ Detailed textural & color information
- ✓ Essential for **surface cracks** & corrosion



## IR Channel

- ✓ **Co-located infrared camera** capture
- ✓ Detects temperature-related anomalies
- ✓ Reveals **subsurface defects** invisible in RGB



## Multispectral Image Fusion

- 1 **Precise pixel-wise alignment** of IR to RGB
- 2 **Homography-based registration** for focal length differences
- 3 Fusion function: weighted sum to deep learning models (IFCNN)

## Fusion Function

$$\text{Fusion}(x,y) = f(\text{RGB}(g_x(x,y), g_y(x,y)), \text{IR}(h_x(x,y), h_y(x,y)))$$

Where  $g_x$ ,  $g_y$  and  $h_x$ ,  $h_y$  represent geometric transformations applied to RGB and IR images, and  $f(\cdot, \cdot)$  is the fusion function

# YOLO Models



## YOLOv8 Baseline

- ✓ **High-speed inference** with state-of-the-art accuracy
- ✓ Ideal balance for **near real-time** UAV inspection
- ✓ Trained on both RGB and fused multispectral images



## Thermal Model (Mt)

- ✓ Exclusively trained on **IR-heavy data**
- ✓ Heightened sensitivity to subtle thermal gradients
- ✓ Detects defects often overlooked by general-purpose models



## Defect Classes



C1

### Cracks

Structural fractures compromising aerodynamic integrity

**Detection:** Visible surface fractures in blades,



C2

### Corrosion

Oxidative degradation on metallic surfaces

**Detection:** Primarily on tower and blade root connections



C3

### Overheating

Temperature anomalies indicating mechanical/electrical faults

**Detection:** Primarily via IR channel in nacelle

# Ensemble Fusion Algorithm



## Bounding Box Fusion

- 1 Identify **overlapping boxes** of same class from both models
- 2 Calculate **weighted averages** of confidence & parameters
- 3 Create fused detection with combined information
- 4 Preserve unique findings from each model
- 5 Apply **Non-Maximum Suppression** to remove redundancies

## Weighted Average Formulas

$$p_{\text{ensemble}} = \gamma p_{\text{Mt},j} + (1-\gamma)p_{y,i}$$

$$x_{\text{ensemble}} = \gamma x_{\text{Mt},j} + (1-\gamma)x_{y,i}$$

$$y_{\text{ensemble}} = \gamma y_{\text{Mt},j} + (1-\gamma)y_{y,i}$$

$$W_{\text{ensemble}} = \gamma W_{\text{Mt},j} + (1-\gamma)W_{y,i}$$

$$H_{\text{ensemble}} = \gamma H_{\text{Mt},j} + (1-\gamma)H_{y,i}$$

Where  $\gamma \in [0, 1]$  controls the contribution of each model



## Weighting Factor

$\gamma = 0.5$  assigns equal importance to both models



## NMS Threshold

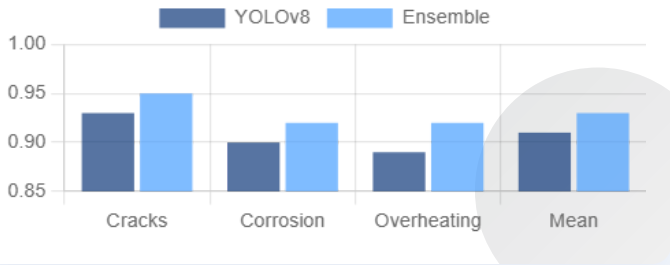
Removes boxes with IoU above threshold  $\tau_{\text{nms}}$

# Results - Performance Comparison

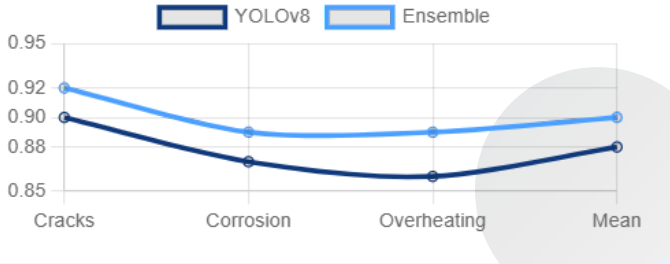
## YOLOv8 vs. Ensemble Performance

| Defect Class     | mAP@.5 |             | mAP@.5:.95 |             | Precision |             | F1-S        |
|------------------|--------|-------------|------------|-------------|-----------|-------------|-------------|
|                  | YOLOv8 | Ensemble    | YOLOv8     | Ensemble    | YOLOv8    | Ensemble    | YOLOv8      |
| Cracks (C1)      | 0.93   | <b>0.95</b> | 0.78       | <b>0.81</b> | 0.91      | <b>0.93</b> | 0.90        |
| Corrosion (C2)   | 0.90   | <b>0.92</b> | 0.75       | <b>0.78</b> | 0.88      | <b>0.90</b> | 0.87        |
| Overheating (C3) | 0.89   | <b>0.92</b> | 0.74       | <b>0.77</b> | 0.87      | <b>0.90</b> | 0.86        |
| Mean             | 0.91   | <b>0.93</b> | 0.76       | <b>0.79</b> | 0.89      | <b>0.91</b> | <b>0.88</b> |

mAP@.5 Comparison



F1-Score Comparison



### Overall Improvement

Ensemble achieves **+2.2%** mAP@.5



### Thermal Defects

Overheating detection shows



### Balanced Performance

Ensemble maintains high precision

# Results - State-of-the-Art Comparison

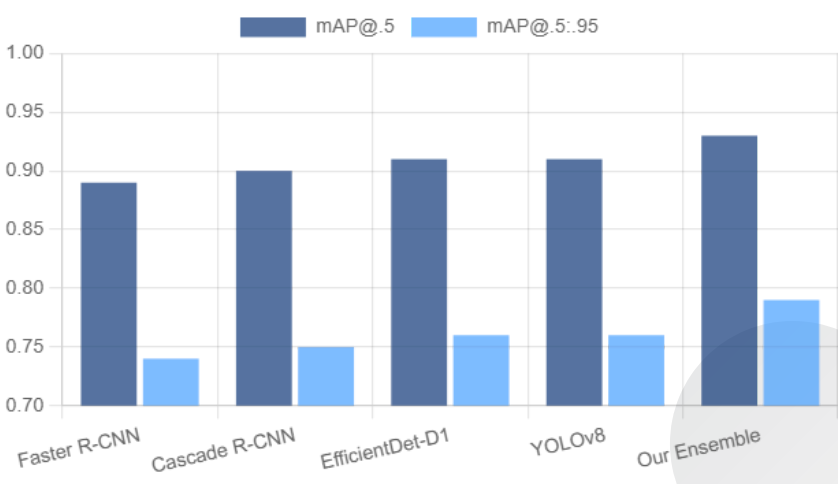


## Performance Comparison

| Model                    | mAP@.5      | mAP@.5:.95  |
|--------------------------|-------------|-------------|
| Faster R-CNN             | 0.89        | 0.74        |
| Cascade R-CNN            | 0.90        | 0.75        |
| EfficientDet-D1          | 0.91        | 0.76        |
| YOLOv8                   | 0.91        | 0.76        |
| Our Ensemble <b>OURS</b> | <b>0.93</b> | <b>0.79</b> |



## Visual Comparison



### Top Performance

Our ensemble achieves **highest mAP@.5 (0.93)** and **mAP@.5:.95 (0.79)** among all models



### Single-Stage Advantage

Single-stage detectors (YOLOv8, EfficientDet) outperform two-stage approaches in speed-accuracy balance



### Ensemble Benefits

Fusion of YOLOv8 with specialized thermal model provides **+3.9%** improvement in stricter mAP@.5:.95 metric

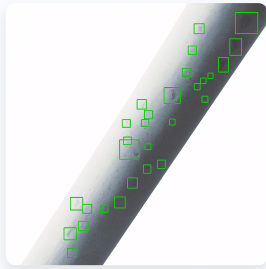
# Results - Qualitative Analysis

Comparison of RGB images vs. fused multispectral images demonstrates enhanced defect visibility

## Blade Cracks

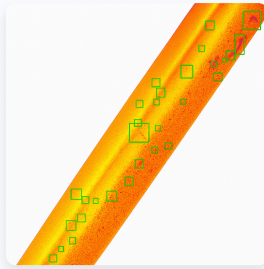
+2.2% mAP

RGB Image



Visibility hampered by shadows

Fused Image



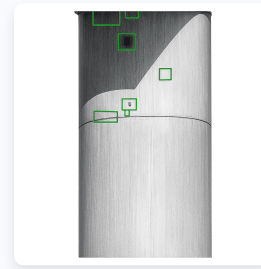
Enhanced fracture line contrast



## Tower Corrosion

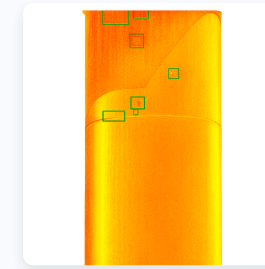
+2.0% mAP

RGB Image



Subtle surface degradation

Fused Image



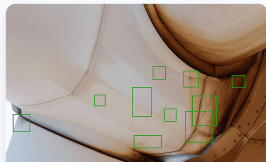
Distinct corrosion patterns



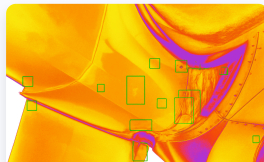
## Rotor Stress

+3.0% mAP

RGB Image



Fused Image



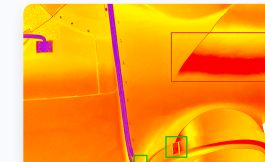
## Motor Overheating

+3.4% mAP

RGB Image



Fused Image



# Conclusion and Future Work



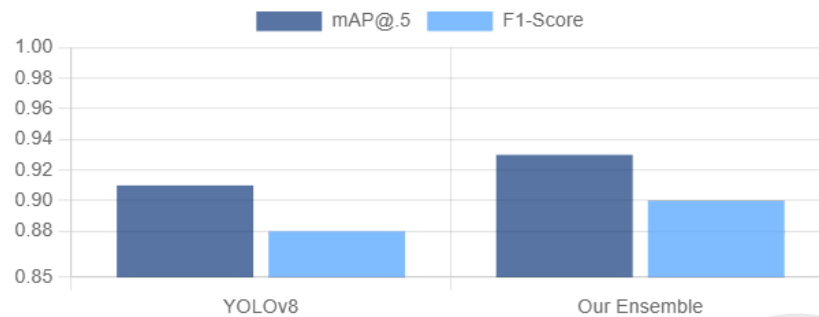
## Key Findings

📈 **mAP@.5: 0.93 | F1-score: 0.90**

↑ **+2.2%** improvement over single YOLOv8

🌡️ Superior **thermal defect** detection

🔍 Enhanced **subtle cracks** & **corrosion** visibility



## Advantages

🔄 **Multi-modal synergy** for comprehensive diagnostics

⚖️ **Speed** of single-stage + **precision** of specialist model

⚠️ Reduces **false negatives** & **false positives**

👁️ Addresses **single-spectral limitations**



## Future Research

🧠 **Edge deployment**  
optimization for UAVs

📷 **Robust sensor fusion**  
algorithms

🌪️ **Environmental noise**

🌈 **Hyperspectral**